aiPHeed: food-insecurity risk indicators for CALABARZON (DA Region IV-A)

Model aiPHeed-E36-LGBM (LightGBM, government indicators + news signals) E36.1. Forecast quarter 2026-Q4, issued at origin 2026-10-01 with information published before it. Record SHA-256 4be4a6202daa2462b84283e8daec5b8cb5c0a5859e627feaea24b170315579f0.
aiPHeed produces potential food-insecurity risk indicators for review. They are not official food-insecurity classifications and do not replace assessments by authorized government agencies.

What is estimated

T1: S1 = H + 2D > tau, HIGH or LOW (the approved Chapter I composite food-stress label)
H: SWS Balance Luzon self-rated hunger (%), one regional value for all five provinces (measured)
D: province food inflation minus the five-province mean (pp), PSA provincial CPI (measured)
Not: an official or measured food-insecurity prevalence; provincial differences come only from prices
Evidence status: EXPERIMENTAL / NOT VALIDATED (no baseline is beaten significantly; development dates only)

CALABARZON: official regional hunger (measured)

Quarter	Round	Hunger %	Posted
2024-Q2	Second Quarter 2024 SWS Survey, June 23-July	19.6	2024-07-23
2024-Q3	September 2024 SWS Survey	18.1	2024-10-17
2024-Q4	December 2024 SWS Survey	25.3	2025-01-14
2025-Q1	Stratbase-SWS March 15-20, 2025 National Surv	24.0	2025-03-29
2025-Q4	November 2025 SWS Survey	16.7	2026-01-14
2026-Q1	March 2026 SWS Survey	22.4	2026-05-12
2026-Q2	June 2026 SWS Survey	17.3	2026-08-24
2026-Q3	September 2026 SWS Survey	23.6	2026-10-14

regional: ONE value shared by all five provinces; not a provincial or municipal measurement

Cavite: 2026-Q4 forecast (lead 1)

	2026-Q4 FORECAST	2026-Q3 backtest (observed HIGH)
P(HIGH)	98%	98%
Signal	STRONG	STRONG

Main drivers (TreeSHAP, log-odds):

Factor	Value	Contribution
Province food-price pressure in the first two months of this quarter	0.06	+0.88
Change in regional hunger between the two latest surveys	-5.10	-0.82
Province food-price pressure last quarter (PSA food CPI)	0.87	+0.64
Latest regional hunger rate (SWS Balance Luzon)	17.30	+0.62
Latest published risk label for the province	1.00	+0.61
Latest regional hunger relative to the HIGH threshold	2.31	-0.51

exact TreeSHAP contribution of each input to this estimate, in log-odds (positive raises the probability of HIGH); associations the model learned from 2021-2026 data, not causes

The model puts Cavite at 98% for 2026-Q4. Raised mainly by: Province food-price pressure in the first two months of this quarter; Province food-price pressure last quarter (PSA food CPI). Lowered mainly by: Change in regional hunger between the two latest surveys.

Food inflation in Cavite was 0.47 pp above the five-province average in 2026-Q3 (PSA, measured).

News evidence (2026-Q3): 0 validated article(s) of 57 province-attributed; status NO_USABLE_EVIDENCE.

Review: the model estimate is high; check provincial market prices and field reports. Prioritise for review by DA Region IV-A; do not allocate resources on this estimate alone.

Top LGUs (relative position)	Risk index	Poverty used	Kind
Gen. Mariano Alvarez	0.623	9.540674	PSA municipal SAE estimate
General Emilio Aguinaldo	0.5966	16.0907	PSA municipal SAE estimate
Cavite City	0.5904	12.708223	PSA municipal SAE estimate
Tagaytay	0.5686	15.283253	PSA municipal SAE estimate
Mendez	0.5483	15.108867	PSA municipal SAE estimate

Laguna: 2026-Q4 forecast (lead 1)

	2026-Q4 FORECAST	2026-Q3 backtest (observed HIGH)
P(HIGH)	99%	100%
Signal	STRONG	STRONG

Main drivers (TreeSHAP, log-odds):

Factor	Value	Contribution
Province food-price pressure in the first two months of this quarter	0.46	+1.25
Change in regional hunger between the two latest surveys	-5.10	-0.77
Province rice-price inflation relative to the regional mean	-8.71	+0.72
Latest published risk label for the province	1.00	+0.60
Latest regional hunger relative to the HIGH threshold	2.31	-0.54
Latest regional hunger rate (SWS Balance Luzon)	17.30	+0.52

exact TreeSHAP contribution of each input to this estimate, in log-odds (positive raises the probability of HIGH); associations the model learned from 2021-2026 data, not causes

The model puts Laguna at 99% for 2026-Q4. Raised mainly by: Province food-price pressure in the first two months of this quarter; Province rice-price inflation relative to the regional mean. Lowered mainly by: Change in regional hunger between the two latest surveys.

Food inflation in Laguna was 0.27 pp above the five-province average in 2026-Q3 (PSA, measured).

News evidence (2026-Q3): 7 validated article(s) of 52 province-attributed; status SUFFICIENT.

Review: the model estimate is high; check provincial market prices and field reports. Prioritise for review by DA Region IV-A; do not allocate resources on this estimate alone.

Top LGUs (relative position)	Risk index	Poverty used	Kind
Siniloan	0.6079	24.356336	PSA municipal SAE estimate
San Pedro	0.4618	5.987922	PSA municipal SAE estimate
Biñan	0.4551	5.750266	PSA municipal SAE estimate
Cabuyao	0.267	5.015342	PSA municipal SAE estimate
Santa Cruz	0.2613	9.748186	PSA municipal SAE estimate

Batangas: 2026-Q4 forecast (lead 1)

	2026-Q4 FORECAST	2026-Q3 backtest (observed HIGH)
P(HIGH)	17%	100%
Signal	STRONG	STRONG

Main drivers (TreeSHAP, log-odds):

Factor	Value	Contribution
Province food-price pressure in the first two months of this quarter	-2.19	-2.24
Latest published risk label for the province	0.00	-1.57
Latest regional hunger relative to the HIGH threshold	2.31	-0.90
Province rice-price inflation relative to the regional mean	-5.41	+0.74
Latest regional hunger rate (SWS Balance Luzon)	17.30	+0.67
Province food-price pressure last quarter (PSA food CPI)	-2.43	-0.67

exact TreeSHAP contribution of each input to this estimate, in log-odds (positive raises the probability of HIGH); associations the model learned from 2021-2026 data, not causes

The model puts Batangas at 17% for 2026-Q4. Lowered mainly by: Province food-price pressure in the first two months of this quarter; Latest published risk label for the province; Latest regional hunger relative to the HIGH threshold.

Food inflation in Batangas was 2.63 pp below the five-province average in 2026-Q3 (PSA, measured).

News evidence (2026-Q3): 0 validated article(s) of 73 province-attributed; status NO_USABLE_EVIDENCE.

Lower estimate: the model expects this province below the threshold; it can still be wrong (12 of 65 development estimates were wrong). Prioritise for review by DA Region IV-A; do not allocate resources on this estimate alone.

Top LGUs (relative position)	Risk index	Poverty used	Kind
Lipa	0.1524	12.027362	PSA municipal SAE estimate
Bauan	0.1267	10.654363	PSA municipal SAE estimate
San Pascual	0.1263	11.372837	PSA municipal SAE estimate
Taal	0.1163	9.453046	PSA municipal SAE estimate
San Jose	0.1059	9.862561	PSA municipal SAE estimate

Rizal: 2026-Q4 forecast (lead 1)

	2026-Q4 FORECAST	2026-Q3 backtest (observed HIGH)
P(HIGH)	98%	98%
Signal	STRONG	STRONG

Main drivers (TreeSHAP, log-odds):

Factor	Value	Contribution
Province food-price pressure in the first two months of this quarter	0.51	+1.17
Change in regional hunger between the two latest surveys	-5.10	-0.82
Latest published risk label for the province	1.00	+0.62
Latest regional hunger rate (SWS Balance Luzon)	17.30	+0.53
Latest regional hunger relative to the HIGH threshold	2.31	-0.52
Relevant food-insecurity news in the region (count)	2.30	-0.44

exact TreeSHAP contribution of each input to this estimate, in log-odds (positive raises the probability of HIGH); associations the model learned from 2021-2026 data, not causes

The model puts Rizal at 98% for 2026-Q4. Raised mainly by: Province food-price pressure in the first two months of this quarter; Latest published risk label for the province. Lowered mainly by: Change in regional hunger between the two latest surveys.

Food inflation in Rizal was 0.71 pp above the five-province average in 2026-Q3 (PSA, measured).

News evidence (2026-Q3): 2 validated article(s) of 40 province-attributed; status LIMITED_SIGNAL.

Review: the model estimate is high; check provincial market prices and field reports. Prioritise for review by DA Region IV-A; do not allocate resources on this estimate alone.

Top LGUs (relative position)	Risk index	Poverty used	Kind
Taytay	0.9807	9.808941	PSA municipal SAE estimate
Antipolo	0.6447	9.294249	PSA municipal SAE estimate
Cardona	0.6274	9.233761	PSA municipal SAE estimate
Morong	0.5796	8.997131	PSA municipal SAE estimate
San Mateo	0.5388	6.644608	PSA municipal SAE estimate

Quezon: 2026-Q4 forecast (lead 1)

	2026-Q4 FORECAST	2026-Q3 backtest (observed HIGH)
P(HIGH)	99%	100%
Signal	STRONG	STRONG

Main drivers (TreeSHAP, log-odds):

Factor	Value	Contribution
Province food-price pressure in the first two months of this quarter	1.16	+1.16
Change in regional hunger between the two latest surveys	-5.10	-0.82
Province food-price pressure last quarter (PSA food CPI)	1.00	+0.64
Latest published risk label for the province	1.00	+0.61
Latest regional hunger rate (SWS Balance Luzon)	17.30	+0.53
Latest regional hunger relative to the HIGH threshold	2.31	-0.52

exact TreeSHAP contribution of each input to this estimate, in log-odds (positive raises the probability of HIGH); associations the model learned from 2021-2026 data, not causes

The model puts Quezon at 99% for 2026-Q4. Raised mainly by: Province food-price pressure in the first two months of this quarter; Province food-price pressure last quarter (PSA food CPI). Lowered mainly by: Change in regional hunger between the two latest surveys.

Food inflation in Quezon was 1.17 pp above the five-province average in 2026-Q3 (PSA, measured).

News evidence (2026-Q3): 0 validated article(s) of 40 province-attributed; status NO_USABLE_EVIDENCE.

Review: the model estimate is high; check provincial market prices and field reports. Prioritise for review by DA Region IV-A; do not allocate resources on this estimate alone.

Top LGUs (relative position)	Risk index	Poverty used	Kind
Lucena	0.6803	16.8	province-mean substitute (no munic
San Francisco	0.6104	30.214238	PSA municipal SAE estimate
Macalelon	0.6071	30.101027	PSA municipal SAE estimate
San Andres	0.5569	27.817577	PSA municipal SAE estimate
Infanta	0.5464	27.335361	PSA municipal SAE estimate

Model evaluation: one quarter ahead, development quarters

E36 frozen protocol c83d11cb2ac8; lead 1; nested rolling-origin; run 20261015T023921Z

Approach	Correct/n	Acc	Acc 95% CI	Bal.acc	Prec	Recall	F1	ROC-AUC	Brier
Proposed: combined LightGBM (government + news), nes	53/65	0.815	0.738-0.892	0.822	0.956	0.811	0.878	0.850	0.154
B1 Naive persistence	57/65	0.877	0.800-0.938	0.860	0.959	0.887	0.922	0.860	0.123
B2 Logistic regression, combined features	56/65	0.862	0.785-0.938	0.818	0.940	0.887	0.913	0.840	0.120
B3 Logistic regression, government indicators only	55/65	0.846	0.754-0.923	0.809	0.939	0.868	0.902	0.866	0.102
B4 Logistic regression, NLP features only	46/65	0.708	0.554-0.846	0.692	0.905	0.717	0.800	0.658	0.198
B5 LightGBM, government indicators only	55/65	0.846	0.769-0.923	0.841	0.957	0.849	0.900	0.870	0.145
B6 LightGBM, NLP features only	44/65	0.677	0.492-0.846	0.673	0.900	0.679	0.774	0.736	0.195
Diagnostic: always HIGH (majority class)	53/65	0.815	0.677-0.923	0.500	0.815	1.000	0.898	0.500	0.185

One-quarter-ahead accuracy of the proposed LightGBM on 65 development estimates (13 quarters, 2023-Q1 to 2026-Q3): 53/65 = 0.815. The point estimate is above 75%, but the 95% date-bootstrap interval (0.738 to 0.892) reaches below 75%, so the target is not demonstrated. Always-HIGH reaches the same accuracy (0.815) and persistence a higher one (0.877). Unlike always-HIGH, the model does identify LOW quarters (balanced accuracy 0.822). No paired difference with any of the six baselines is significant. These are development dates; the 2026-Q4 forecast is the first untouched test.

News relevance validation (blind holdout)

Rule R1: precision 0.852, recall 0.395, F1 0.54.
Data sufficiency (news, FSSI-v2): {"collected": 8320, "province_attributed": 2820, "relevant_R1": 140, "syndicated_copies_removed": 0, "admitted_to_FSSI_v2": 140, "province_quarters_sufficient_v2": 7, "province_quarters_limited_v2": 46, "province_quarters_none_v2": 67, "in_FSSI_v1": 55, "province_quarters_sufficient_v1": 0}

City and municipal ranking

Indicative, relative ordering of the 142 cities and municipalities for planning attention. It is the province estimate re-weighted by PSA 2021 poverty incidence (60%) and 2024 POPCEN population density (40%) inside each province. It is not a measured municipal food-insecurity rate, not an independent municipal forecast and not an official classification (Chapter I, 1.7). The province estimate it starts from has NOT been validated against baselines (E25-E28), so the ranking inherits that status: EXPERIMENTAL.
- It does not say that any LGU is food insecure, or how many households are.

- Positions inside a province reflect the vulnerability weights only. Positions across provinces are not a measured comparison: the vulnerability index is scaled 0 to 1 inside each province, so the region-wide order mixes five different scales and follows the within-province index more than the province estimate (E35-D4). Compare LGUs within their province.
- For an LGU without a PSA municipal poverty estimate (Lucena City, a highly urbanized city that the PSA no-HUC table excludes), the poverty term is a province substitute, so its position inside the province comes from population density and the substitute value, not from a municipal measurement.
- Two LGUs with equal risk_index share a rank; their listing order (PSGC) carries no meaning.

Limitations

- The label is dominated by one regional SWS hunger value, so in most quarters all five provinces share a label; 53 of 65 development labels are HIGH. Provincial differences come mainly from relative food prices.
- One-quarter-ahead accuracy is 0.815 on 65 development estimates, the same as always-HIGH and below persistence (0.877). The 75% target is met as a point estimate but not demonstrated (95% interval reaches 0.738). No baseline is beaten significantly.
- News signals added no measurable skill: the government-only LightGBM (B5) scores as well or better than the combined model on the same rows.
- News evidence is title-only (Google News RSS), thin in older quarters, and its relevance labels were made by the AI assistant; a researcher re-review is pending.
- LGU rankings re-weight the province forecast by PSA 2021 municipal poverty (141 of 142 LGUs; Lucena City, a highly urbanized city absent from the PSA no-HUC table, uses a province substitute of unverified source) and 2024 POPCEN density. They are relative positions, not municipal forecasts.
- The 2026-Q4 forecast is the first untouched test; it can be scored only after the SWS December 2026 round is released.

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